Network Troubleshooting, Management, Monitoring, and Optimization

Task 1 : Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

Command Used:

ipconfig /all

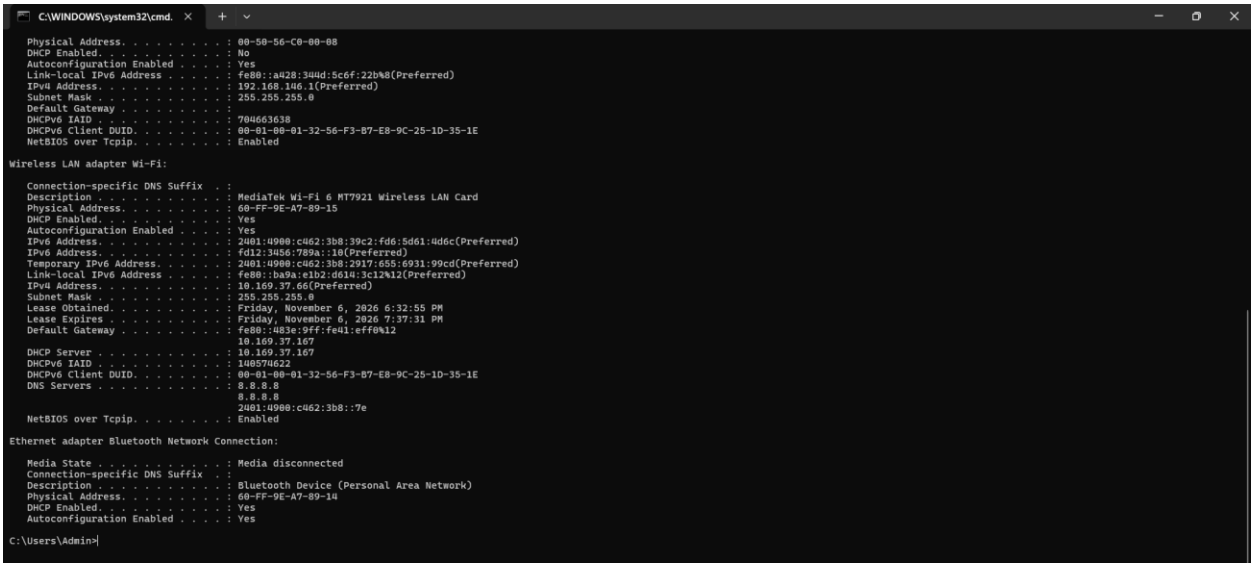
Procedure:

1. Open Command Prompt.
2. Type ipconfig /all and press Enter.
3. Find the Wireless LAN adapter Wi-Fi section.
4. Identify the IPv4 address, subnet mask, default gateway, and DNS servers.

Result:

- IPv4 Address: 10.169.37.66
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.169.37.167
- Primary DNS Server: 8.8.8.8
- Secondary DNS Server (IPv6): 2401:4900:c462:3b8::7e

Screenshot:



Task 2 : Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Tool Used: Windows Command Prompt

Command Used:

ping www.google.com

Procedure:

1. Open Windows Command Prompt.
2. First, run the ping command while connected to Wi-Fi.
3. Note the replies and average response time.
4. Disconnect the Wi-Fi connection and run the same command again.
5. Observe the error message displayed.
6. Reconnect Wi-Fi and repeat the ping command to verify connectivity.

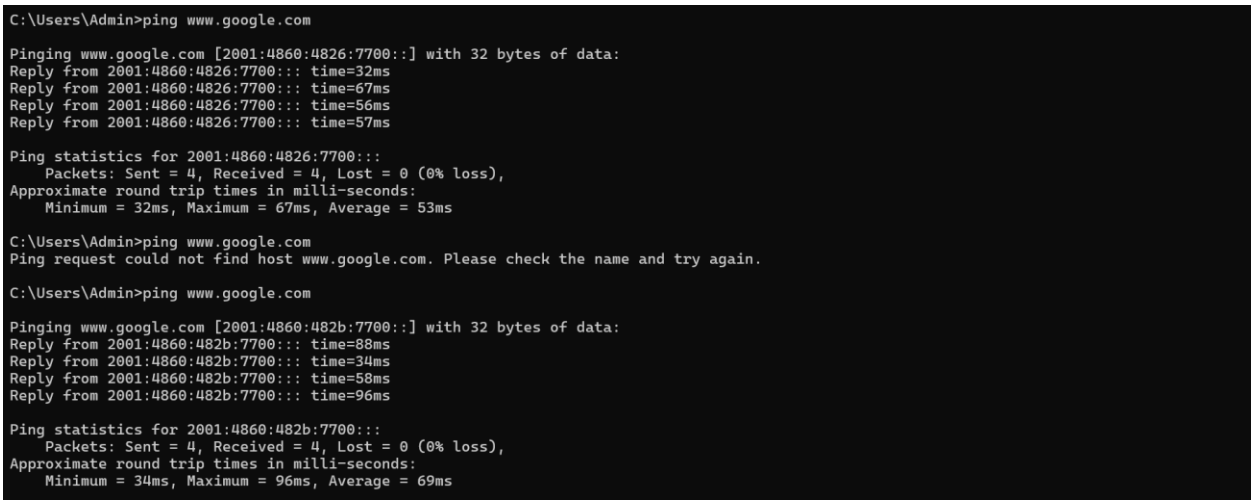
Result:

Network Status	Result	Average Response Time
Wi-Fi Connected	Successful	53 ms
Wi-Fi Disconnected	Failed	N/A
Wi-Fi Reconnected	Successful	69 ms

Observations:

- **Wi-Fi Connected:** All 4 packets were received successfully with 0% packet loss. The average response time was 53 ms.
- **Wi-Fi Disconnected:** The command displayed "Ping request could not find host www.google.com", indicating that the domain name could not be resolved.
- **Wi-Fi Reconnected:** The computer successfully reached Google's server again. All 4 packets were received with 0% packet loss and an average response time of 69 ms.

Screenshot:



```
C:\Users\Admin>ping www.google.com

Pinging www.google.com [2001:4860:4826:7700::] with 32 bytes of data:
Reply from 2001:4860:4826:7700::: time=32ms
Reply from 2001:4860:4826:7700::: time=67ms
Reply from 2001:4860:4826:7700::: time=56ms
Reply from 2001:4860:4826:7700::: time=57ms

Ping statistics for 2001:4860:4826:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 32ms, Maximum = 67ms, Average = 53ms

C:\Users\Admin>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Users\Admin>ping www.google.com

Pinging www.google.com [2001:4860:482b:7700::] with 32 bytes of data:
Reply from 2001:4860:482b:7700::: time=88ms
Reply from 2001:4860:482b:7700::: time=34ms
Reply from 2001:4860:482b:7700::: time=58ms
Reply from 2001:4860:482b:7700::: time=96ms

Ping statistics for 2001:4860:482b:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 34ms, Maximum = 96ms, Average = 69ms
```

Task 3 : Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

To monitor network activity on a Windows computer while streaming a YouTube video and identify which process consumes the most bandwidth.

Procedure:

1. Open YouTube in Google Chrome and play a video continuously.
2. Press Ctrl + Shift + Esc to open Task Manager.
3. Navigate to the Performance tab and select Wi-Fi.
4. Monitor the network usage graph for 5 minutes.
5. Observe the peak network usage during video streaming.
6. Open the Processes tab and check the Network column to identify the application consuming the most bandwidth.

Observation:

Parameter	**Result**
Network Adapter	MediaTek Wi-Fi 6 MT7921
Connection Type	Wi-Fi (802.11n)
Monitoring Duration	5 Minutes
Peak Network Usage	Approximately 7.7 Mbps
Application Used for Streaming	Google Chrome
Current Network Usage	0 Mbps (at screenshot time)

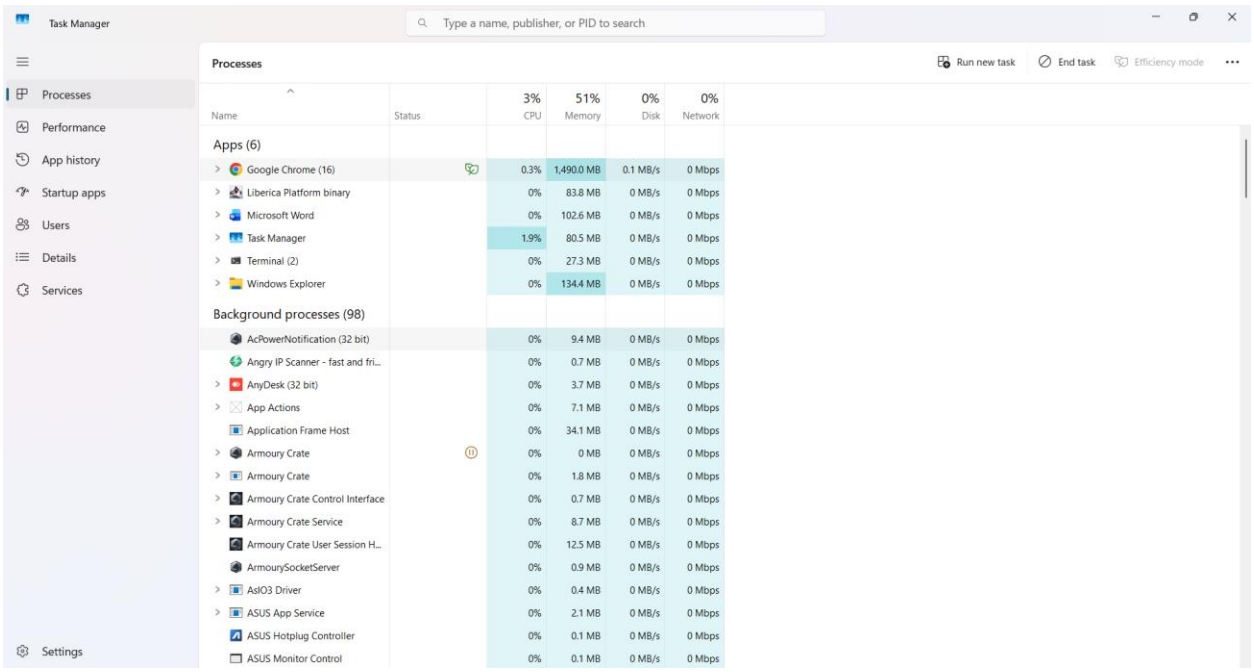
Result:

During the monitoring process, network activity increased while streaming the YouTube video. The Wi-Fi performance graph showed a peak network usage of approximately 7.7 Mbps.

Google Chrome was used to stream the video. Network usage became 0 Mbps when no active data transfer was taking place.

Screenshot:


Task Manager
Performance
Run new task
CPU
1% 3.76 GHz
Memory
8.0/15.7 GB (51%)
Disk 0 (C: D:)
SSD (NVMe)
0%
Ethernet
VMware Network Ad...
S: 0 R: 0 Kbps
Ethernet
VMware Network Ad...
S: 0 R: 0 Kbps
Wi-Fi
Wi-Fi
S: 0 R: 0 Kbps
GPU 0
Intel(R) UHD Graphics
0% N/A
GPU 1
NVIDIA GeForce RTX...
0% (45 °C)
Wi-Fi
MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
Throughput
11 Mbps
7.7 Mbps
60 seconds
0
Send
0 Kbps
Receive
0 Kbps
Adapter name: Wi-Fi
SSID: :0(:):&):
Connection type: 802.11n
IPv4 address: 10.169.37.66
IPv6 address: 2401:4900:c462:3b8:39c2:fd6:5d61:4d6c
Signal strength:


Task Manager
Type a name, publisher, or PID to search
Processes
Run new task
End task
Efficiency mode
Processes
Performance
App history
Startup apps
Users
Details
Services
Settings
Name
Status
3% CPU
51% Memory
0% Disk
0% Network
Apps (6)
Google Chrome (16) 0.3% 1,490.0 MB 0.1 MB/s 0 Mbps
Liberica Platform binary 0% 83.8 MB 0 MB/s 0 Mbps
Microsoft Word 0% 102.6 MB 0 MB/s 0 Mbps
Task Manager 1.9% 80.5 MB 0 MB/s 0 Mbps
Terminal (2) 0% 27.3 MB 0 MB/s 0 Mbps
Windows Explorer 0% 134.4 MB 0 MB/s 0 Mbps
Background processes (98)
AcPowerNotification (32 bit) 0% 9.4 MB 0 MB/s 0 Mbps
Angry IP Scanner - fast and fri... 0% 0.7 MB 0 MB/s 0 Mbps
AnyDesk (32 bit) 0% 3.7 MB 0 MB/s 0 Mbps
App Actions 0% 7.1 MB 0 MB/s 0 Mbps
Application Frame Host 0% 34.1 MB 0 MB/s 0 Mbps
Armoury Crate 0% 0 MB 0 MB/s 0 Mbps
Armoury Crate 0% 1.8 MB 0 MB/s 0 Mbps
Armoury Crate Control Interface 0% 0.7 MB 0 MB/s 0 Mbps
Armoury Crate Service 0% 8.7 MB 0 MB/s 0 Mbps
Armoury Crate User Session H... 0% 12.5 MB 0 MB/s 0 Mbps
ArmourySocketServer 0% 0.9 MB 0 MB/s 0 Mbps
AsIO3 Driver 0% 0.4 MB 0 MB/s 0 Mbps
ASUS App Service 0% 2.1 MB 0 MB/s 0 Mbps
ASUS Hotplug Controller 0% 0.1 MB 0 MB/s 0 Mbps
ASUS Monitor Control 0% 0.1 MB 0 MB/s 0 Mbps

Task 4 : Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

To capture network traffic while browsing Instagram Web and analyze HTTP/HTTPS packets using Wireshark.

Procedure:

1. Downloaded and installed Wireshark on the Windows computer.
2. Opened Wireshark and selected the active Wi-Fi network interface.
3. Started capturing network packets.
4. Opened Instagram Web in Google Chrome and browsed the website.
5. Captured network traffic for more than 30 seconds.
6. Stopped the packet capture and applied the display filter to examine secure web traffic.
7. Observed the source IP, destination IP and protocol information in the captured packets.

Display Filter Used:

http or tls

Observations:

During packet analysis, the following information was identified:

1. **Source IP Address:** Identifies the device sending the network packets.
2. **Destination IP Address:** Identifies the remote server receiving the packets.
3. **Protocol Information:** Shows the communication protocol, such as TLS 1.2 and TLS 1.3, used for secure communication.

The capture also showed QUIC packets and the hostname gateway.instagram.com.

Result:

Wireshark successfully captured network traffic while browsing Instagram Web. The captured packets showed encrypted communication between the computer and remote servers.

Task 5 : Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

To generate a troubleshooting checklist using an AI tool to improve home network speed and reliability for online gaming.

AI-Generated Response:

If a friend is experiencing slow internet, high ping or lag while playing online games, the following steps can help improve network performance:

1. Use a Wired Ethernet Connection

Connect the gaming PC or console directly to the router using an Ethernet cable. A wired connection is generally more stable and provides lower latency than Wi-Fi.

2. Restart the Router

Restart the Wi-Fi router to refresh the network connection and resolve temporary connectivity issues.

3. Reduce Background Network Usage

Close unnecessary applications, downloads and streaming services running in the background because they can consume internet bandwidth.

4. Place the Router Properly

Keep the router in a central and elevated location. Avoid placing it near thick walls or electronic devices that may interfere with Wi-Fi signals.

5. Use the 5 GHz Wi-Fi Band

Connect to the 5 GHz Wi-Fi band when possible. It generally provides faster speeds and less interference over shorter distances.

6. Update Network Drivers and Router Firmware

Keep the computer's network adapter drivers and router firmware updated to improve network stability and fix connectivity issues.

7. Check Ping and Packet Loss

Use the ping command to check response time and packet loss. High ping or packet loss can cause delays and interruptions during online gaming.

Suggestion I Would Try First:

I would first try using a wired Ethernet connection instead of Wi-Fi because it generally provides a more stable connection, reduces wireless interference and helps improve gaming performance.

Conclusion:

AI-generated troubleshooting suggestions can help identify common network problems and improve internet speed, stability and reliability. Following these steps can provide a better online gaming experience.